

THAPAR TRADE

A Campus Marketplace for Student-to-Student Trading

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1. Introduction and Problem Statement

Student-to-student exchange of used products is a common requirement within a university environment. At Thapar, students may need to buy or sell textbooks, calculators, cycles, electronics, furniture, hostel appliances and other academic or personal items. These transactions are currently facilitated largely through informal channels such as WhatsApp groups, ION groups, personal contacts and word of mouth.

Although these channels enable communication, they do not provide the functionality of a structured marketplace. Product information becomes distributed across multiple conversations, making relevant listings difficult to discover. Buyers cannot reliably search or filter products by category, price, condition or availability. Sellers may also fail to update or remove listings after an item has been sold, resulting in unnecessary communication and inaccurate availability information.

A further challenge is the reliability of marketplace listings. For example, a student may jokingly or misleadingly list a product belonging to another student as if it were their own, even though the actual product is not for sale. A general marketplace does not provide a campus-specific mechanism through which such listings can be reported and reviewed.

Thapar Trade proposes a centralized, campus-specific marketplace that addresses these limitations through structured product listings, student authentication, search and filtering, availability management, reporting and administrative moderation. Students will be able to report listings that they believe are unauthorized, misleading or falsely claim ownership. Reported listings can then be reviewed by an administrator and removed or restored as appropriate.

The project therefore aims to make student-to-student trading more organized, searchable and reliable while providing a practical mechanism for handling misleading or unauthorized listings.

2. Objectives

The primary objective is to design and implement a functional campus marketplace that provides a structured workflow for student-to-student trading.

The project will pursue the following objectives:

- Develop an authenticated web platform through which students can create, browse and manage product listings.
- Implement keyword-based search and filtering by category, price, condition and availability.
- Implement listing states of *Available*, *Reserved* and *Sold* to maintain accurate product availability.
- Provide structured product information including descriptions, prices, conditions and images.
- Implement a reporting mechanism for suspicious, misleading or unauthorized listings.
- Develop an administrative moderation interface for reviewing reports and taking appropriate action.
- Validate the system through functional testing and user-acceptance testing with a small group of student users.

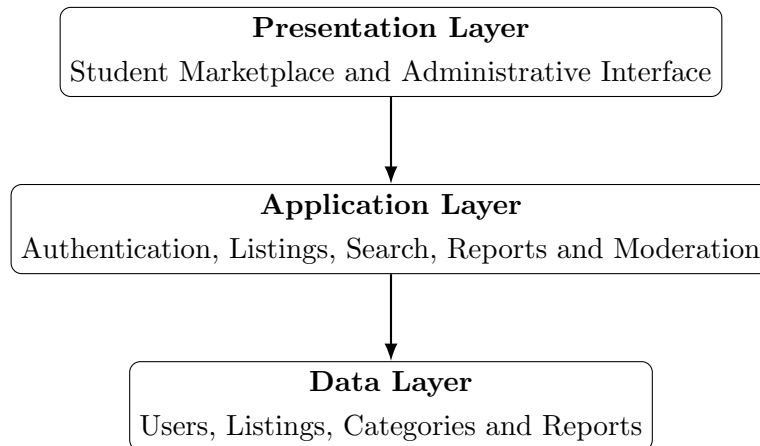
The project will prioritize a complete and reliable core marketplace before considering advanced functionality.

3. Methodology

The project will follow an iterative software-engineering methodology in which the system is developed as a set of integrated functional modules. The methodology will cover architecture design, module implementation, system integration, testing and validation.

3.1. System Architecture

The proposed system will use a three-layer architecture consisting of a presentation layer, application layer and data layer.



The presentation layer will provide responsive interfaces for students and administrators. The application layer will implement authentication, listing management, search, reporting and moderation logic. The data layer will maintain structured records for users, listings, categories and reports.

The final implementation technologies will be selected according to team familiarity, development requirements and availability of appropriate free or academic-tier services.

3.2. Functional Modules

The system will be implemented through the following modules:

- **Authentication and User Management:** The platform will associate marketplace activity with an authenticated student account.
- **Product Listing:** Sellers will provide standardized information including product title, category, description, price, condition and images. Required fields will be validated before publication.
- **Marketplace Search:** Students will be able to search listings using keywords and refine results using category, price, condition and availability filters.
- **Listing Lifecycle:** Sellers will manage product availability using *Available*, *Reserved* and *Sold* states.
- **Product Details and Seller Information:** Buyers will be able to view product information, listing condition, price and relevant seller information before contacting the seller.
- **Reporting Mechanism:** Users will be able to report listings that they believe are misleading, unauthorized, inappropriate or falsely claim ownership.
- **Administrative Moderation:** Administrators will review submitted reports and may temporarily restrict, remove or restore listings based on the review.

3.3. Reporting and Moderation Workflow

The reporting mechanism is designed to address cases in which a student lists another student's product without authorization, including misleading or prank listings.

A user can select "Report Listing" from the product page and choose a relevant reason, such as:

- Product is not for sale.
- I am the actual owner.
- Seller is falsely claiming ownership.
- Product information is misleading.
- Other.

The report will be stored in the database and made available to the administrator through the moderation dashboard.

The workflow will be:

Listing → Report → Administrative Review → Restore or Remove

This approach does not require every student to register their personal possessions in advance. Instead, it provides a practical mechanism for identifying and responding to disputed or misleading listings.

3.4. System Execution and Integration

Development will proceed incrementally, beginning with the database schema and authentication layer, followed by product listing and marketplace functionality. Search, filtering and listing-status management will then be integrated with the core marketplace.

The principal workflow will be:

**Authentication → Create Listing → Publish → Search → Contact Seller →
Reserved/Sold**

The reporting and moderation workflow will operate independently alongside the marketplace and can be triggered whenever a user identifies a potentially problematic listing.

3.5. Testing and Validation

Testing will be conducted at feature, integration and user-acceptance levels.

Individual functions such as authentication, listing creation, search, status updates and reporting will first be tested independently. Integrated workflows will then be tested to ensure that data remains consistent between the user interface, backend and database.

A small group of approximately 10–20 students will participate in user-acceptance testing. Representative tasks will include:

- Locating a product using search and filters.
- Creating and publishing a product listing.
- Updating a listing from Available to Sold.
- Reporting a suspicious or unauthorized listing.
- Reviewing a reported listing through the administrative interface.

Evaluation will consider task completion, errors, time required and user feedback. The findings will be used to refine the interface and workflow before the final demonstration.

3.6. Constraints and Scope Control

The project will focus on a realistic semester-scale prototype. The system will not claim to automatically verify physical ownership of arbitrary products. It will instead provide authenticated student identities, seller declarations, reporting and administrative moderation.

Payment processing, delivery services, automated fraud detection, AI-based ownership detection and complex recommendation systems are outside the initial scope. Advanced features may be considered only after the core marketplace is stable.

4. Timeline

Weeks	Development Phase	Expected Deliverable
1–2	Requirements analysis and system design	Requirements, user flows, database design and UI prototype
3–4	Authentication and backend foundation	Working authentication and core backend services
5–7	Marketplace and listing module	Create, edit, browse and view product listings
8–9	Search and listing management	Search, filters and Available/Reserved/Sold workflow
10	Reporting and administrative moderation	Report submission, moderation dashboard and listing actions
11	Integration and user testing	Integrated prototype and student evaluation
12	Refinement and buffer	Final testing, documentation and presentation

Table 1: Proposed semester development timeline

The final phase includes a buffer period for unexpected development, integration or testing delays.

5. Resources and Budget

The project will be developed by the three-member team. Work will be distributed across frontend development, backend and database development, interface design, testing, documentation and system integration. Team members will collaborate during integration and validation.

Required resources include student laptops, a code editor, Git/GitHub for version control, a database, web-development frameworks and an appropriate deployment environment. Product images will require suitable storage during development and testing.

No specialized laboratory equipment is required. Free or academic-tier development and hosting services will be preferred where available. Consequently, the project is expected to require no significant external expenditure.

Resource	Expected Cost
Development tools and code editor	No additional cost
Git/GitHub	No additional cost
Database services	Free/academic tier where available
Hosting/deployment	Free/academic tier where available
Testing devices	Existing student devices

Table 2: Resources and estimated project cost

No external funding is required for the proposed academic prototype.